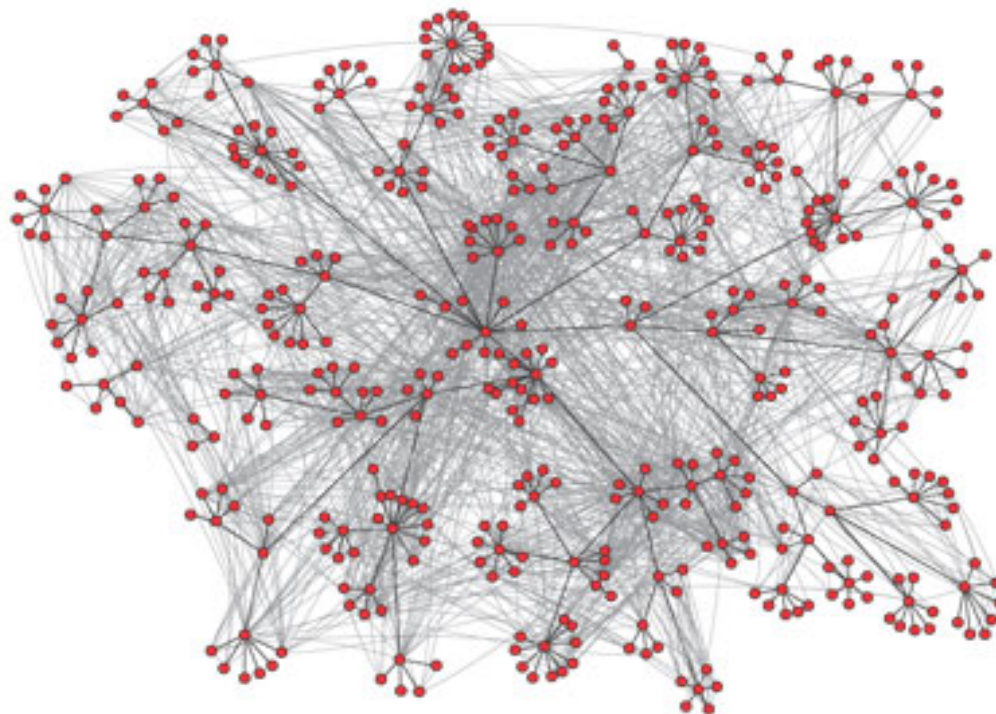


Starter Topic: Structure of Graphs

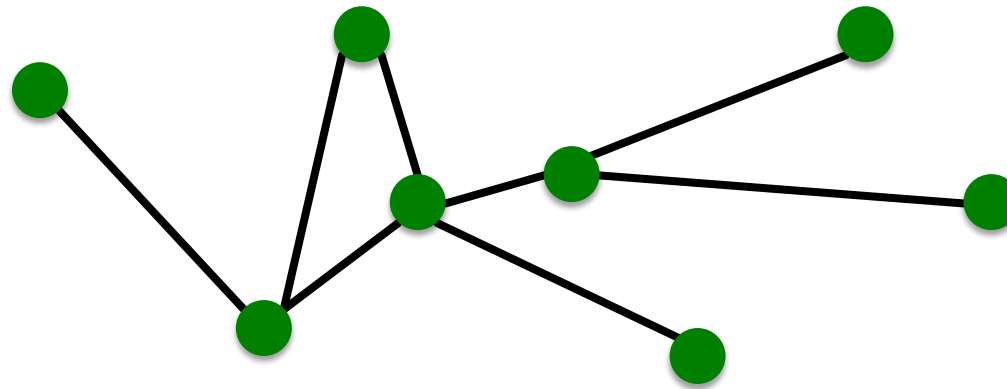
Structure of Networks?



A network is a collection of objects where some pairs of objects are connected by links

What is the structure of the network?

Components of a Network



- **Objects:** nodes, vertices
- **Interactions:** links, edges
- **System:** network, graph

N

E

$G(N,E)$

Networks or Graphs?

- **Network** often refers to real systems
 - Web, Social network, Metabolic network

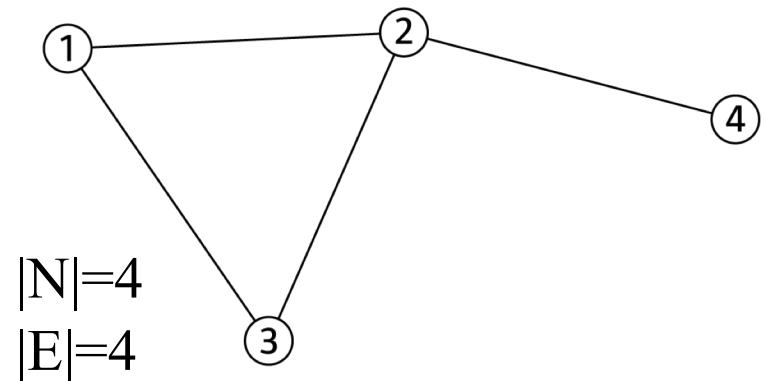
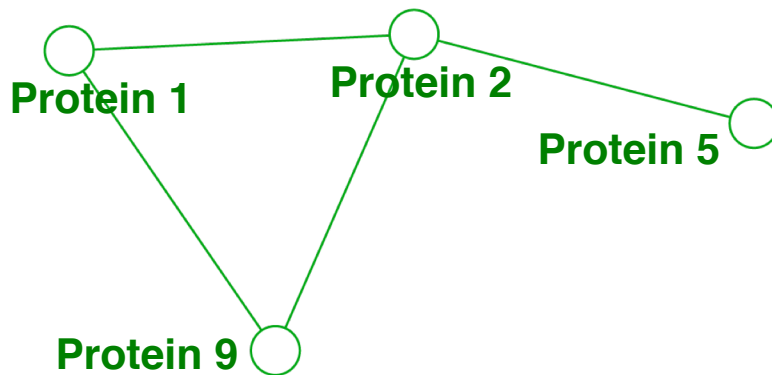
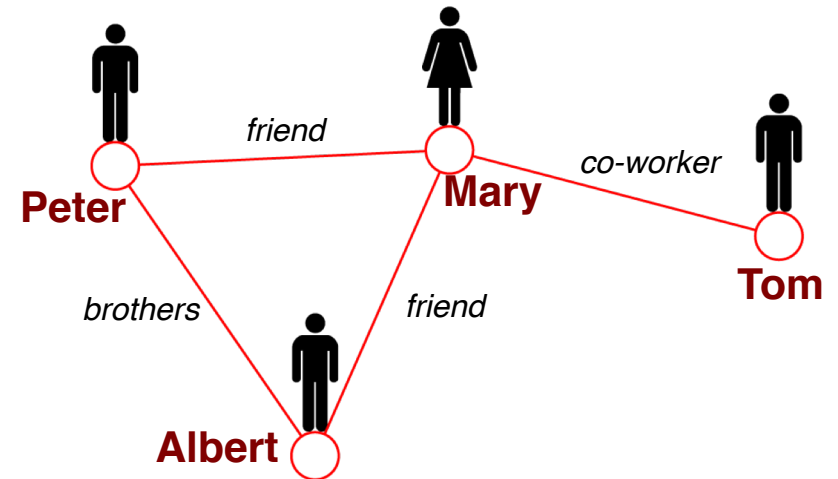
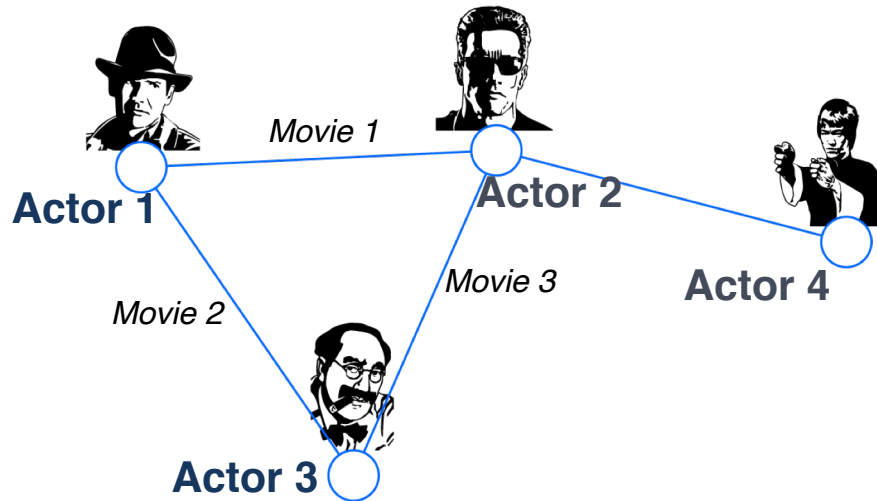
Language: Network, node, link

- **Graph** is a mathematical representation of a network
 - Web graph, Social graph (a Facebook term)

Language: Graph, vertex, edge

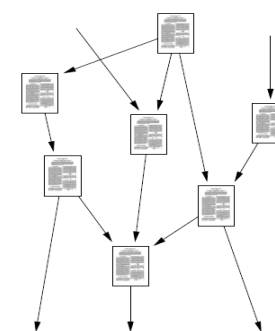
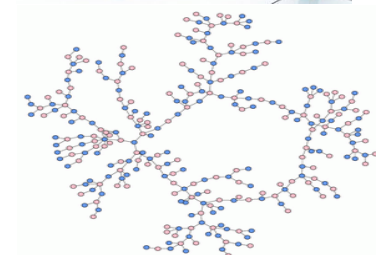
We will try to make this distinction whenever it is appropriate, but in most cases we will use the two terms interchangeably

Networks: Common Language



Choosing Proper Representations

- If you connect individuals that work with each other, you will explore a **professional network**
- If you connect those that have a sexual relationship, you will be exploring **sexual networks**
- If you connect scientific papers that cite each other, you will be studying the **citation network**
- **If you connect all papers with the same word in the title, what will you be exploring?** It is a network, nevertheless



How do you define a network?

- **How to build a graph:**
 - What are nodes?
 - What are edges?
- **Choice of the proper network representation of a given domain/problem determines our ability to use networks successfully:**
 - In some cases there is a unique, unambiguous representation
 - In other cases, the representation is by no means unique
 - The way you assign links will determine the nature of the question you can study